

# Production Grade RAG Python App

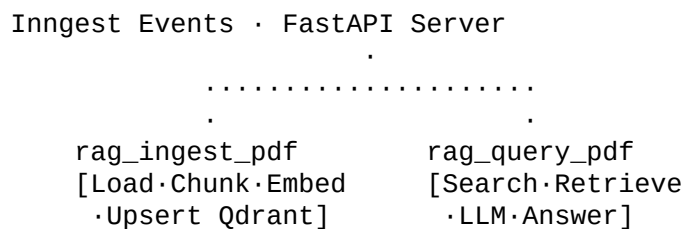
A **production-ready RAG application** using **Inngest** for serverless workflow management and **Qdrant** for vector storage · with built-in throttling and rate limiting.

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## · Features

- **PDF Ingestion** · Automated load, chunk, embed, upsert pipeline
  - **Vector Search** · Qdrant semantic search
  - **LLM Generation** · OpenAI-powered contextual answers
  - **Production Controls** · Throttle (2/min) and rate limit (1/4hr)
  - **Event-Driven** · Inngest serverless function orchestration
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## .. Architecture



## · Getting Started

```
cd ProductionGradeRAGPythonApp
uv sync
uv run python main.py
```

## Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
QDRANT_URL=...
QDRANT_API_KEY=...
```

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## · Logic Flow

### Ingestion (throttled)

- 1 PDF loaded and text extracted
- 2 Text chunked into segments
- 3 Chunks embedded via OpenAI
- 4 Vectors upserted to Qdrant

### Query

- 1 User question embedded
  - 2 Top-k similar chunks retrieved
  - 3 Context assembled from chunks
  - 4 OpenAI generates contextual answer
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## · Dependencies

`fastapi, ingest, langchain, qdrant-client, openai, python-dotenv`

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## · License

Educational project · use freely for learning and reference.